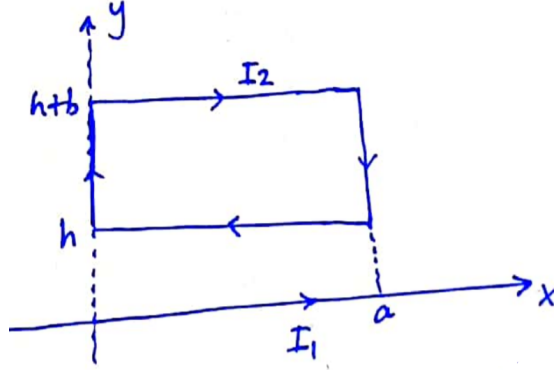


Solucionario pc6 - Física 3

1. Sea el sistema



Espira:

- i) $x = 0, h \leq y \leq h + b$
- ii) $y = h + b, 0 \leq x \leq a$
- iii) $x = a, h + b \geq y \geq h$
- iv) $y = h, a \geq x \geq 0$

Fuerza magnético: $\vec{F} = \int i d\vec{r} \times \vec{B}$; campo del alambre: $\vec{B}(y) = \frac{\mu_0 I_1}{2\pi y} \hat{k}$

$$\text{i) } d\vec{r} = dy \hat{j} : \vec{F}_1 = I_2 \int_h^{h+b} dy \hat{j} \times \left(\frac{\mu_0 I_1}{2\pi y} \hat{k} \right) = \frac{\mu_0 I_1 I_2}{2\pi} \ln \left(\frac{b+h}{h} \right) \hat{i}$$

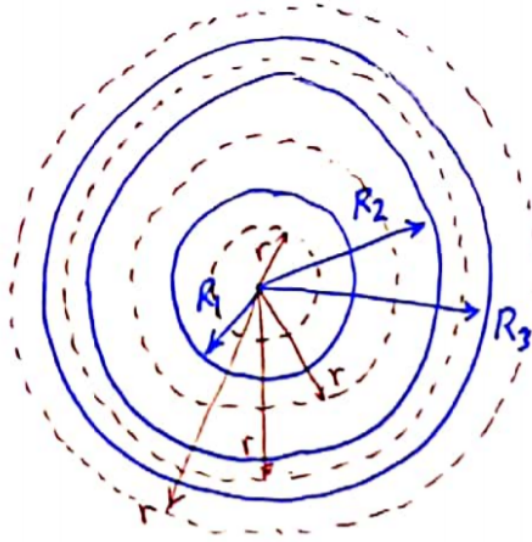
$$\text{iii) } d\vec{r} = -dy \hat{j} : \vec{F}_3 = -I_2 \int_h^{h+b} dy \hat{j} \times \left(\frac{\mu_0 I_1}{2\pi y} \hat{k} \right) = -\vec{F}_1$$

$$\text{ii) } d\vec{r} = dy \hat{i} : \vec{F}_2 = I_2 \int_0^a dy \hat{i} \times \left(\frac{\mu_0 I_1}{2\pi (b+h)} \hat{k} \right) = -\frac{\mu_0 I_1 I_2 a}{2\pi (b+h)} \hat{j}$$

$$\text{iv) } d\vec{r} = -dx \hat{i} : \vec{F}_4 = -I_2 \int_0^a dx \hat{i} \times \left(\frac{\mu_0 I_1}{2\pi h} \hat{k} \right) = \frac{\mu_0 I_1 I_2 a}{2\pi h} \hat{j}$$

$$\vec{F}_R = \vec{F}_1 + \vec{F}_2 + \vec{F}_3 + \vec{F}_4 = \frac{\mu_0 I_1 I_2 a}{2\pi} \left(\frac{1}{h} - \frac{1}{b+h} \right) \hat{j} \quad (\text{Fuerza de Repulsión}).$$

2. Sea la siguiente figura:



(a) cilindro interior: $r < R_1$

$$\int \vec{B} \cdot d\vec{l} = \mu_0 I_C$$

$$\begin{aligned} \pi R_1^2 &\rightarrow I \\ \pi R^2 &\rightarrow I_C \end{aligned}$$

$$\rightarrow I_C = I \left(\frac{r^2}{R_1^2} \right)$$

$$B(2\pi r) = \mu_0 I \cdot \left(\frac{r^2}{R_1^2} \right) \rightarrow B = \frac{\mu_0 I r}{2\pi R_1^2} = (15 r) \text{ mT} \quad (\text{r en cm})$$

- Entre los conductores: $R_1 < r < R_2$:

$$\int \vec{B} \cdot d\vec{l} = \mu_0 I_C \rightarrow B(2\pi r) = \mu_0 I \rightarrow B = \frac{\mu_0 I}{2\pi r} = \left(\frac{2.4}{r} \right) \text{ mT}, \quad \text{r en cm.}$$

- conductor exterior: $R_2 < r < R_3$:

$$\int \vec{B} \cdot d\vec{l} = \mu_0 I_C \rightarrow B(2\pi r) = \mu_0 [I + (-I_{ext})],$$

$-I_{ext}$: corriente que fluye por el conductor exterior. (sentido contrario)

$$I_{ext} = I \cdot \left(\frac{A_C}{A} \right) = I \cdot \left(\frac{\pi R_1^2 - \pi R_2^2}{R_3^2 - R_2^2} \right)$$

$$B(2\pi r) = \mu_0 \left[I - I \left(\frac{R_1^2 - R_2^2}{R_3^2 - R_2^2} \right) \right] = \mu_0 I \left(\frac{R_3^2 - R^2}{R_3^2 - R_2^2} \right) \rightarrow B = \frac{\mu_0 I}{2\pi r} \left(\frac{R_3^2 - R^2}{R_3^2 - R_2^2} \right)$$

$$B = 3.16 \left(\frac{4 - r^2}{r} \right) \text{ mT.} \quad (\text{r en cm})$$

- cilindro exterior: $\int \vec{B} \cdot d\vec{l} = \mu_0 I_C = 0 \rightarrow B=0$.

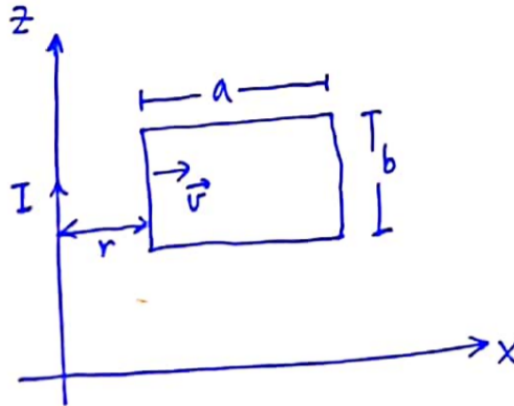
(b) • cilindro interior: $H = \frac{B}{\mu} = \frac{B}{\mu_0 \mu_r} = \frac{I_r}{2\pi \mu_r R_1^2}, \mu_r = 600 \rightarrow H \approx (20 r) T \quad (r: \text{cm})$

- Entre conductores: $H = \frac{B}{\mu \mu_r} = \frac{I}{2\pi \mu_r r}, \mu_r = 1 \rightarrow H \approx \left(\frac{1900}{r}\right) T, \quad (r: \text{cm})$

- conductor exterior: $H = \frac{B}{\mu_0 \mu_r} = \frac{I (R_3^2 - r^2)}{2\pi \mu_r (R_3^2 - R_2^2) r}, \mu_r = 1200$
 $H \approx 2 \left(\frac{4 - r^2}{r}\right) T, \quad r \text{ en cm.}$

- cilindro exterior: $H=0$.

3. Sea el siguiente gráfico.



Campo producido por el alambre infinito:

$$\vec{B}(x) = \frac{\mu_0 I}{2\pi x} \hat{j}$$

Flujo magnético:

$$\phi = \int \vec{B} \cdot d\vec{S} = \int_0^b \int_r^{r+a} \frac{\mu_0 I}{2\pi x} \hat{j} \cdot dx dz \hat{j}$$

$$\rightarrow \phi = \frac{\mu_0 I b}{2\pi} \ln \left(\frac{r+a}{r} \right), \quad \text{pero } r = r(t) = h + vt \rightarrow \phi = \frac{\mu_0 I b}{2\pi} \ln \left(\frac{h + vt + a}{h + vt} \right)$$

Ley de Faraday: $\varepsilon = -\frac{d\phi}{dt} = -\frac{\mu_0 I b}{2\pi} \left[\frac{-av}{(h + vt)(h + vt + a)} \right]$

$$\rightarrow \varepsilon = \frac{\mu_0 I a b}{2\pi (h + vt)(h + vt + a)} v$$

Por la ley de Lenz, la corriente en el circuito debe ser horaria.

4. $I = 20 \text{ mA}; \psi_L = 200; \mu_r = 1200$

(a) $B = \mu_0 \mu_r \left(\frac{NI}{L} \right) = (4\pi \times 10^{-7} \times 1200 \times 2000 \times 20 \times 10^{-3}) = 0.06 \text{ T}$

(b) Sin el núcleo de hierro: $B = \mu_0 (NI/L) \rightarrow I = \frac{B}{\mu_0 (N/L)} = \frac{0.06}{4\pi \times 10^{-7} \times 2000} = 23,9 \text{ A}.$